

Abdelrahman Osama

Computer Engineering Student

Cairo, Egypt | reach.abdelrahman@gmail.com | (+20)-15-5000-6367

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

German University in Cairo (GUC)

Bachelor of Computer Engineering | **GPA: 1.42 / 0.7 (German Scale)**

Cairo, Egypt

Expected May 2028

Dar Jana International Schools (DJIS)

American Diploma | **GPA: 4.0 / 4.0**

Jeddah, Saudi Arabia

Jun 2023

Technical & Creative Skills

Programming Languages: Java, Python, JavaScript, HTML, CSS, MySQL

Data & Analytics: Google Sheets, Excel, Microsoft Dynamics 365

Creative & Media: DaVinci Resolve, Blender, Adobe Photoshop

Core Competencies: KPI Dashboards Creation, GUI Development, Object-Oriented Programming

Professional Experience

SoftTrend

IT Solutions Marketing

Cairo, Egypt

July 2025 – Aug 2025

- Facilitated B2B sales for enterprise software and cloud solutions, including Microsoft 365 and endpoint protection.
- Consulted clients on infrastructure needs, recommending hardware and network security setups like Fortinet firewalls.
- Drafted technical proposals and marketing materials to communicate the ROI of integrated IT systems.

German University in Cairo (GUC)

Computer Science Jr. Teaching Assistant

Cairo, Egypt

March 2025 – Present

- Taught undergraduate Computer Science lab sessions, explaining core programming concepts and algorithm design.
- Evaluated practical assignments and provided feedback to reinforce fundamental data structures.
- **Courses Supported:** CSEN403 (Concepts of Programming Languages), CSEN401 (Computer Programming Lab), CSEN202 (Introduction to Computer Programming), and CSEN102 (Introduction to Computer Science).

Technical Projects

Interactive demos and detailed architectures for all projects are available on [my portfolio](#).

Operating System Simulator & Visualizer | *Java, Swing, Multithreading*

- Built a multi-threaded OS simulator with a custom interpreter and Memory Management Unit (MMU) to allocate resources within a 40-word RAM array.
- Programmed a memory swapping engine that serializes dormant process data to disk, preventing failures at 100% memory capacity.
- Implemented Round Robin, HRRN, and MLFQ scheduling algorithms with cross-queue preemption for high-priority tasks.
- Created Mutex objects to manage hardware locks, dynamically routing processes to blocked queues to prevent deadlock and starvation.
- Developed a Java Swing UI to visualize real-time execution, queue states, and dynamically generated Gantt charts for CPU metrics.

HR Management System | *C#, ASP.NET Core 8.0, SQL Server, IIS*

- Built an HR management dashboard in ASP.NET Core 8.0 and C# using MVC architecture for server-side routing and data handling.
- Deployed the application to a cloud-based IIS environment, configuring environment variables and remote SQL Server connection strings.
- Resolved deployment blocks by bypassing active server file-locks and debugging production-level SQL execution errors.
- Secured domain integrity via SSL/HTTPS protocols and verified site ownership with Google Search Console.

Jackaroo Board Game Application | *Java, JavaFX, Eclipse*

- Led a development team to build a digital Jackaroo board game, managing task delegation and code integration.
- Designed the backend state management using MVC architecture and core OOP principles (Polymorphism, Inheritance).
- Built an interactive JavaFX GUI with custom animations mapped to backend game logic events.
- Programmed independent probability algorithms to manage a 102-card deck and validate complex user moves.

Automated Finance & Analytics Dashboards | *Google Sheets*

- Created a cloud-based personal finance tracker to automate mobile transaction categorization.
- Built a central analytics dashboard to aggregate transactional data across 18 specialized tabs.
- Designed B2B sales templates to track revenue sources, transaction statuses, and individual team performance.

Leadership & Extracurriculars

IEEE’25 Junior Technical Program

Cairo, Egypt

Media & Design Team Member

Oct 2024 – May 2025

- Directed video production for technical content, increasing exposure to Web Development concepts for high school students.
- Designed graphic assets and branding materials to guide students through university life events and announcements.

Additional Information

Languages: Arabic (Native), English (C1 - IELTS 7.5), Spanish (A2), German (A1)

Standardized Testing: SAT Score 1480/1600 (Math: 800)